

RAM AND STORAGE THEORY

1.Create a comparison table listing the main differences between DDR3, DDR4, and DDR5 RAM, including speed, voltage, and typical use cases.

ANS:

Feature	DDR3	DDR4	DDR5
1.Speed	800-2133 MT/S	1600-3200 MT/S	4800-8800+MT/S
2.Voltage	1.5v(1.35v For DDR3L)	1.2v	1.1v
3.Typical Use Cases	laptops,Older PCs and legacy systems	Everyday computers,gaming PCs and workstations	High performance gaming ,content creation ,Ai,and latest PCs

2.Open your laptop or desktop (with power disconnected), locate the RAM slots, and take a clear photo of the installed RAM module. Reinstall the RAM module and verify that your system boots up successfully.

Hint: Refer to your device manual or a trusted YouTube guide if unsure about RAM removal and installation steps.

ANS:

Safety & Preparation

Power Down: Completely shut down your computer.

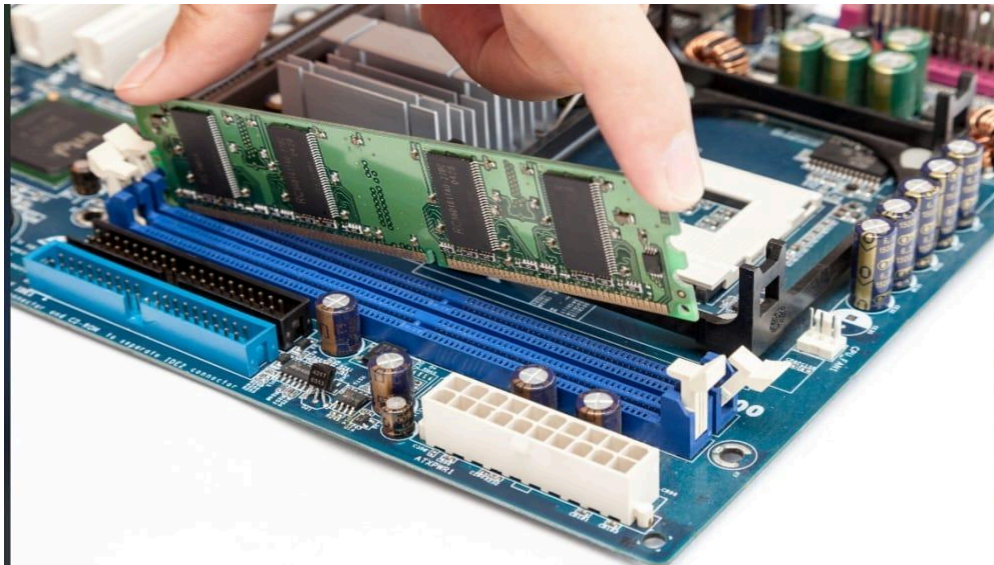
Disconnect Power: Unplug the power cable (desktop) or remove the battery if accessible (laptop).

Clear Static: Press and hold the power button for 5 seconds to drain any residual power. Touch a metal part of your computer case (or an unpainted metal object) to ground yourself and prevent static discharge.

2. Locate and Remove the RAM

Open the Case: Remove the side panel (desktop) or bottom access door/plate (laptop).

Locate the Slots: Look for long vertical or horizontal slots near the CPU.



Release the Module:

Desktop: Push down on the plastic clips at both ends of the RAM slot. The RAM stick will pop up slightly.

Laptop: Gently pull the metal retention arms on both sides outward. The module will tilt up at an angle.

Remove: Hold the RAM stick by its side edges and pull it straight out. Avoid touching the gold contacts.

3. Reinstall and Verify

Align the Notch: Look at the bottom of the RAM stick—there is a small notch off-center. Align this notch with the ridge inside the RAM slot so it only goes in one way.

Insert & Lock:

Desktop: Push the module firmly straight down into the slot until the side clips snap back up into place automatically.

Laptop: Insert the module at a 30-degree angle, then press down flat until the side clips click into place.

3. List three key differences between HDD, SSD, and NVMe storage types, and for each, name one app or use case (from apps you use daily) that would benefit most from that storage type.

ANS:

Feature	HDD	SATA SSD	NVMe SSD
Technology	Mechanical: Use spinning magnetic platter and a moving read/write head.	Flash memory: Uses NAND flash chips with no moving parts over a SATA interface.	Flash Memory: Uses NAND flash chips running over high-speed PCIe lanes.
Speed	Slow: read/write speeds typically hover around 80-160MB/s.	Moderate: Capped by the SATA connection at around 500-550MB/s.	Ultra-fast: reaches speeds from 3500 MB/s to over 7500MB/s
Durability & Size	Bulkier & Fragile: Sensitive to physical shocks and drops; larger 3.5" or 2.5" form factor.	Durable & Compact: Resistant to drops; comes in 2.5" drives or older M.2 sticks.	Durable & Extremely Small: Resistant to drops; uses tiny M.2 gum-stick form factors directly on the motherboard.

Daily Apps & Ideal Use Cases

1. HDD: Mass Media Archiving

Ideal App: Plex or File Explorer / Finder (for local media libraries).

Why: HDDs offer the lowest cost per gigabyte, making them ideal for storing massive libraries of movies, music, photos, or long-term backups where transfer speeds aren't critical.

2. SATA SSD: Everyday General Computing

Ideal App: Google Chrome or Microsoft Word.

Why: SATA SSDs are significantly faster than HDDs, providing near-instant boot times for your operating system and rapid launch times for daily web browsing, productivity apps, and standard software without needing premium PCIe hardware.

3. NVMe SSD: Heavy Data Transfer & High-End Gaming
Ideal App: DaVinci Resolve (or Adobe Premiere Pro).

Why: Editing raw 4K/8K video requires scrubbing through massive video files simultaneously. An NVMe drive prevents playback stutter and drastically cuts down video rendering and project export times. It is also ideal for modern games with heavy asset streaming.

4. Remove the storage drive (HDD or SSD) from your system, take a photo of the connector type (SATA, M.2, or NVMe), and reinstall the drive. Confirm that your device recognizes the storage after reassembly.

Constraint: Only perform this task if you have access to a device you are allowed to open. If not, find and share a YouTube video link showing the process for your laptop/PC model.

ANS:

HDD:

Mechanical & slow (~150 MB/s). Best for massive media archives (e.g., Plex).

SATA SSD: Flash memory & medium speed (~550 MB/s). Best for daily browsing & apps (e.g., Chrome).

NVMe SSD: High-speed PCIe & ultra-fast (3500+ MB/s). Best for heavy 4K editing & gaming (e.g., DaVinci Resolve).

Connector Identifiers:

SATA (2.5"): L-shaped plastic plug with cables.

M.2 SATA: Small card with 2 notches at the bottom pins.

M.2 NVMe: Small card with 1 notch at the bottom pins.